

Howard (Hao) Ye

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[LinkedIn](#) — [GitHub](#) — [Portfolio](#)

SUMMARY

Results-driven DevOps/DevSecOps Engineer with hands-on experience building automated CI/CD pipelines, secure containerized microservices, and resilient cloud infrastructures. Adept at infrastructure automation (Ansible), application security (RBAC, JWT, rate limiting), and deploying full-stack, AI-integrated platforms. Proven ability to bridge software engineering and IT operations to deliver scalable, production-ready solutions.

TECHNICAL SKILLS

CI/CD & Automation: GitHub Actions, Jenkins, Ansible, Terraform

Containers & Cloud: Docker, Kubernetes, AWS (EC2, RDS, S3), GCP, Nginx, Render, Vercel

DevSecOps & Security: RBAC, JWT, SSH Hardening, UFW, Rate Limiting, API Key Auth

Programming: TypeScript, JavaScript, Python, Node.js, C/C++, React, Vue.js

Databases: PostgreSQL, Redis, SQLite

AI/ML & Testing: PyTorch, Hugging Face, Google Gemini API, Git, Jest, Supertest

EDUCATION

MS Computer Engineering — University of Tennessee Knoxville — 2025

EXPERIENCE

Volunteer AI Software Developer — MatchingDonors.com (Remote) Jan 2026 – Present

Platform: AI-Powered Healthcare App with CI/CD Pipeline — [Repo] — [Demo]

- Building a full-stack organ donor matching platform (React/TypeScript, Node.js) featuring AI-powered article search, real-time Socket.IO chat, and automated Resend API notifications.
- Implemented enterprise-grade CI/CD: GitHub Actions workflow for automated backend testing; Jenkins pipeline with multi-stage checkout, dependency installation, and QA test execution.
- Enforced Role-Based Access Control (RBAC) with JWT authentication and wrote comprehensive integration tests covering admin security scenarios.
- Integrated Google Gemini API for content crawling, AI-driven article categorization, and a conversational chatbot assistant, utilizing SQLite for lightweight data persistence.

PROJECTS

Clinical Data Reconciliation Engine — AI-Powered Healthcare API — [Repo] [Demo]

Tech: Node.js/TypeScript, React, Docker, SQLite, Google Gemini 2.5 Flash, Jest, Render

- Designed and deployed an AI-powered medication reconciliation REST API that resolves conflicting records from multiple healthcare systems (EHR, pharmacy, patient portal) with confidence scoring (0–100%)
- Containerized the full-stack application with Dockerfile and Docker Compose; deployed to Render.com with health check endpoint, environment variable management, and persistent SQLite via Docker volume
- Implemented data quality validation across 4 clinical dimensions (completeness, accuracy, timeliness, plausibility) with severity classification and an overall quality score
- Added production hardening: LLM response caching (~60% cost reduction), rate limiting (30 req/15 min), API key authentication, and full audit trail logging in SQLite; wrote 5 Jest/Supertest unit tests

Cloud-Native HPC Simulation Platform — [Repo]

Tech: Docker, Node.js/TypeScript, PostgreSQL, Redis, AWS

- Architected 4-service microservices platform with Docker Compose orchestration and health checks
- Deployed production-ready application on AWS EC2 with security groups and monitoring capabilities
- Built RESTful API gateway with JWT authentication, Redis caching, and PostgreSQL query optimization
- Implemented comprehensive testing suite covering 15+ API endpoints with automated validation workflows

HPC Cluster Performance Simulator (MS Research Core) — [Repo]

Tech: C/C++, OMNeT++, Distributed Systems, Lustre

- Developed a discrete-event simulation framework using C++ and OMNeT++ to model large-scale HPC storage networks
- Simulated complex Fattree topologies and parallel file systems (Lustre) to evaluate performance bottlenecks
- Engineered automated benchmarking tools and data analysis pipelines for distributed computing environments

Smart Home Energy Management System — [Repo] — [Demo]

Tech: Docker, Vue.js/TypeScript, Node.js, Python, PPO (Reinforcement Learning), Socket.IO

- Built event-driven microservices with 4 containerized services processing real-time data from 11+ IoT devices
- Implemented multi-environment deployment: local Docker development and Vercel production deployment
- Integrated reinforcement learning service with REST APIs for automated device scheduling optimization

Applied AI/ML Techniques Across Kaggle Challenges

Tech: PyTorch, Hugging Face Transformers, XGBoost, LightGBM, CatBoost, Optuna, scikit-learn

- **LLM Response Classification:** Fine-tuned DeBERTa-v3-base transformer for multi-class classification; custom TRL training pipeline with weighted F1-score optimization via hyperparameter tuning
- **Risk Prediction System:** Ensemble gradient boosting with polynomial feature engineering; Optuna hyperparameter search over 100+ trials
- **Bioinformatics Classification:** Multi-label protein function prediction using k-mer extraction and Gene Ontology hierarchy propagation

ACADEMIC EXPERIENCE

MS Computer Engineering — University of Tennessee — 2025

- **Distributed Systems & HPC:** Researched file system performance optimization using machine learning, designing performance modeling frameworks and applying DevOps practices (Docker orchestration, CI/CD).